COMPUTING THE LOWER SHRIEK UP TO SHEAFIFICATION AND TENSOR WITH THE STRUCTURE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the pi-equation fixing the notation; the proof takes the resolution remark for granted and verifies the two assumptions in turn.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The change-of-rings criterion for comparing lower shrieks. Cohomology on Sites, Lemma sites-cohomology-lemma-O-homology-qis, used at proof lines 938-940 and 947-949. Reduces the claim to the two assumptions verified in the proof.
- The resolution remark producing a suitable cosimplicial object. Remark remark-resolution, used at proof lines 941-943. Supplies the cosimplicial object U_bullet from a resolution.
- The computation of Lpi_! of the structure sheaf. used at proof lines 944-945. Gives Lpi_!O = B.
- The computation of Lpi_! of the constant sheaf B. used at proof lines 946-947. Verifies the second assumption of the criterion.

Source reference. Original equation 08PR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11034–11034. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original remark 08QI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11061–11061. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08RX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11110–11110. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $A \to B$ be a ring map. Let π, $\mathcal{O}$, $\underline{B}$ be as in (08PR). For any $\mathcal{O}$-module $\mathcal{F}$ we have*

$$L\pi_!(\mathcal{F}) = L\pi_!(Li^*\mathcal{F}) = L\pi_!(\mathcal{F} \otimes^{\mathbf{L}}_{\mathcal{O}} \underline{B})$$

in $D(Ab)$.

Proof. It suffices to verify the assumptions of Cohomology on Sites, Lemma 08RX hold for $\mathcal{O} \to \underline{B}$ on $\mathcal{C}_{B/A}$. We will use the results of Remark 08QI without further mention. Choose a resolution $P_\bullet$ of B over A to get a suitable cosimplicial object $U_\bullet$ of $\mathcal{C}_{B/A}$. Since $P_\bullet \to B$ induces a quasi-isomorphism on associated complexes of abelian groups we see that $L\pi_!\mathcal{O} = B$. On the other hand $L\pi_!\underline{B}$ is computed by $\underline{B}(U_\bullet) = B$. This verifies the second assumption of Cohomology on Sites, Lemma 08RX and we are done with the proof. □